TO: Professor Sajeela Yaqub

FROM: Shemar Anglin

DATE: April 3, 2025

SUBJECT: Review Article Reflection Memo

Audience & Target Journal

The target journal for my review article is the Journal of Young Investigators. The audience for this journal is made up of mostly undergraduate student researchers and individuals who are early in their careers. Readers are students in the STEM field, careers and professional mentorsm and others who are involved with scientific works. This journal looks for work from those who are credible and developed in a way that those who are not in the related field(s) can understand easily.

Writing Process

My writing process for this assignment started with my desired topic and collection of sources that aligned with what I wanted to discuss. I focused on looking for scholarly articles that discussed artificial intelligence (AI) and the Internet of Things (IoT) within the healthcare industry. After I gathered my sources, I developed my annotated bibliography and received positive feedback. I structured an outline for how I would like the flow of my article to be, more specifically within the discussion of the research. I constructed my paper one section at a time, using the respective sources that I chose for each section. I discussed research, historical background(s), and the innovations of these two technology areas in healthcare.

Feedback & Revisions

I received a lot of feedback from both the instructional team and my peers who were assigned to me. The process of making revisions was a challenge, though it was clear by everyone that I did a very good start with my article. I did learn that I needed to strengthen my use of specific examples, more importantly, in the section that discussed the current research. I added some real-world statistics from my research that I gathered, added some definitions for clarity and referenced specific implementations addressed from my sources. I also needed to ensure that I used citations because I did my research and made my article but I missed points where I needed to cite. For the final draft, it can be made clear that I added some specific examples and definitions for readers to be able to follow and understand the flow and information of this paper. I also made sure that I paid attention to what I was writing and cited appropriately.

Resources & Self-Assessment

I used sources from academic collections and databases that are credible to support every part of the review article. I also use the format of APA 7 and the Journal for Young Investigators. I learned that I have strengths in structuring an organization when it comes to discussion of certain topics. I have gotten better at using what I learned from multiple sources and using hat tof formulate what I need to say. I learned that I need to ork on being more specific and straight to the point with my writing. I need to work on being less broad and formulate a a better analysis. If I were to give myself a letter grade, it would be a C.

I pledge on my honor that I have not given or received any unauthorized assistance on this assignment. – Shemar Anglin | April 3, 2025

Technology in Healthcare: Artificial Intelligence and the Internet of Things (IoT)

Abstract

The healthcare industry has been growing and experiencing various changes due to the increasing use of Artificial Intelligence (AI) and the Internet of Things (IoT). From the advanced AI-based decision-making implementations for early disease detection and personalized care for patients (Bagheri, et al., 2024; Hou, et al., 2022), to the IoT smart devices that regulate and monitor remote health tracking and diagnostics to improve the efficiency and functionality of hospitals and for healthcare professionals (Kumar, et al., 2023; Ahmad, 2024). These technological developments have resulted in faster and more accurate medical conclusions and analyses. But some challenges come from these advancements. From what can be gathered from previous research and analyses involving AI and IoT in healthcare, these benefits do come with challenges, such as ethical and regulatory concerns, security risks of personal patient data, and system compatibility and interoperability constraints (Bajwa, Munir, Nori, & Williams, 2021). AI and IoT technologies will continue to grow and become more advanced, but so will the various concerns. But as of right now, the future looks bright with more technological advancements as the challenges become more manageable and easier to resolve through various innovations and regulation developments.

Introduction

Modern healthcare has shifted significantly due to the integrations of Artificial Intelligence (AI) and the Internet of Things (IoT). AI consists of different algorithms and machine learning implementations to mirror human-based decision-making and problem solving methods, while IoT consists of networks of physical smart devices that collect, analyze, and share data between each other and other systems (Bagheri et al., 2024; Kumar et al., 2023). There are smarter diagnostics from many information systems, improved treatment strategies for patients with various health conditions, and advanced patient monitoring. AI's complex capabilities have played an important role in early disease detection and prediction, diagnosis, and unique personalized treatments for patients through the utilization of machine learning and big data to aid medical professionals in making appropriate decisions (Jimma, 2023). In parallel, IoT has radically changed the capabilities of remote patient monitoring and day-to-day hospital operations, which has allowed healthcare providers to track the vitals and overall health in real time through different kinds of wearable devices and advanced interconnected hospital systems (Sujith, Sajja, Mahalakshmi, Nuhmani, & Prasanalakshmi, 2022). All of the technological advancements that these two areas of technology have led to greater efficiency and better patient outcomes, but concerns are growing in regard to bias within AI-based algorithms within various systems, patient data privacy risks, and concerns, and legal implications that may have a role in how these devices will be regulated.

This review article is for the Journal of Young Investigators, and will cover two important aspects of AI and IoT in healthcare: how AI-based decision-making impacts medical diagnostics and treatments; how IoT devices are strengthening remote monitoring and the overall efficiency of healthcare. Additionally, this paper will explore how these two technologies intersect with each other to contribute to the impacts of healthcare proficiency. Then, there will be a discussion on how these technologies and their innovations face various challenges and ethical concerns.

Background: The Evolution of AI and IoT in Healthcare

Over many decades, the integration of Artificial Intelligence (AI) and the Internet of Things (IoT) within healthcare has grown and evolved into what we know and see now. It has transformed medical decision-making and proper patient care. The first appearances of AI, more specifically, AI in healthcare, date back to the 1970s and the 1980s, when the development of expert systems came into being. MYCIN was one of the first systems to be developed, which was a rule-based expert system with the purpose of aiding in the diagnosing and treatments of various bacterial infections (Bagheri, et al., 2024). However, there were limitations on its potential due to computational and digital medical record constraints. There was a lack of such resources. But hope wasn't lost because real change and growth began in the early 2000s from the increase in machine learning algorithms and shift towards electronic health records (EHRs), which gave the necessary layout for AI-driven healthcare applications and systems (Bagheri, et al., 2024).

With Artificial Intelligence (AI), a major milestone that was achieved at some point was the development of IBM Watson. An amazing system that showed AI's ability to analyze large amounts of medical literature and other forms of data. This system also aided healthcare professionals with medical clinical decision-making. Likewise, AI-driven robotic-assisted surgeries, like the Da Vinci Surgical System, highlighted the potential of AI in reducing surgical errors as much as possible while also improving surgical precision (Bajwa, Munir, Nori, & Williams, 2021).

Similar to AI's growth and evolution, the Internet of Things (IoT) became a fundamental contributor for remote patient monitoring and automation of different hospital processes. The transformation of healthcare through IoT started with wearable devices, such as heart rate monitors and/or step counters in the early 2010s. Then this grew into more advanced technologies that can constantly track blood sugar levels, detect abnormalities within patient heart rhythms, and alert healthcare providers in real time. When the pandemic of COVID-19 came to being, IoT became even more essential for remote patient care and telemedicine (Kelly, Campbell, Gong, & Scuffham, 2020).

These technological developments have enhanced healthcare proficiency through constant patient monitoring, predictive analytics, and early medical action. This decreases the strain on hospitals while ensuring that there are improved patient outcomes (Kelly, Campbell, Gong, & Scuffham, 2020).

As AI and IoT grow, their joint potential in improving healthcare processes and operations, patient experience, and workforce shortages grows as well. We can see this today in the field. However, some concerns continue to grow as well from a technological and regulatory stand point.

Current Research

AI in Medical Diagnostics and Treatment

Artificial Intelligence (AI) integrations in medical diagnostics and treatment have reshaped modern healthcare by providing enhanced accuracy of machine learning, speed, and personalized patient care. Research of today is focused on AI-driven disease detection, enhanced predictive analytics for patient treatment(s), and the ethical concerns regarding AI decision-making. Machine learning algorithms, especially from deep learning models, have shown significant prospects in disease detection, such as cervical cancer, heart disease, and even neurological disorders (Jimma, 2023; Bagheri, et al., 2024). However, to emphasize once again, there are concerns of bias in AI models and legal implications that contribute as challenges/constraints for a larger adoption (Elendu, et al., 2023).

A prominent drive in AI-driven diagnostics within healthcare, or overall, is the use of deep learning models for early disease detection within patients. Deep learning models are a type of machine learning implementation that consist of the use of neural networks that aim to reflect the learning processes of humans (Hou, et al., 2022). An example of this is the convolutional neural networks (CNNs). These systems have been used for cervical cancer screening, which is known to have demonstrated high accuracy in the detection(s) of precancerous lesions in medical images (Hou, et al., 2022). Similarly, there are AI-powered computer-aided diagnostic (CAD) systems that have been most effective in helping radiologists by reducing the events of human errors, and improving the early detection of cancer and cardiovascular diseases that may not be easy to spot by normal non-AI methods (Nagavelli, Samanta, & Chakraborty, 2022).

Another important sector of AI applications is the area of predictive analytics. This is where machine learning algorithms have the ability to analyze large sets of data to be able to make predictions of disease progression and even recommend different treatment and procedure plans. As demonstrated from recent research, it’s proven that AI-driven detection models, focused on heart disease, have improved diagnostic accuracy significantly. This is done by identifying certain patterns in electrocardiograms (ECGs) that human physicians may overlook and not notice until it’s a point where countermeasure may not be too effective or effective at all (Bagheri, et al., 2024).

Nevertheless, these advancements have shown great progress, but AI adoptions in healthcare have raised concerns of ethics, bias, and patient data privacy. One key concern that we have is that there is bias in algorithms, where AI models are operating on limited or unfair data. This can produce inaccurate results and negatively impact certain groups of people who are a part of this technology's utilization (Elendu, et al., 2023). For example, it was discovered that an AI algorithm undermined the severity of Black patients' illnesses compared to the illnesses of white patients; this represents bias that is present within most AI implementations and training data (Elendu, et al., 2023).

In summary, AI is shown to have great potential in bettering patient diagnostics, improving efficiency, and precision in the practice of medicine. However, the ethical challenges must be considered before there can be a full integration into mainstream medical practice within the healthcare industry. Research is constantly being done to discover ways to reduce the occurrence of bias and set standard guidelines for the implementation of AI in healthcare.

IoT in Remote Patient Monitoring & Healthcare Efficiency

The Internet of Things (IoT) within healthcare has provided constant monitoring, improved efficiency within hospitals, and enhanced treatment capabilities for patients. Ongoing research is focused on how wearable IoT devices, implementations of automated hospital systems, and AI-enhanced IoT networks all play a crucial role in healthcare efficiency (Kumar, et al., 2023; Kelly, Campbell, Gong, & Scuffham, 2020). Growing research supports the suggestion that IoT devices are the cause of the decrease of patients being readmitted into hospitals and improving real-time patient monitoring. But, similar to AI, there are challenges with interoperability, patient privacy, and data security. (Kumar, et al., 2023)

A prominent focus for IoT technologies and implementations is remote patient monitoring. This technology consists of many features, where wearable and even implantable devices can track the vital signs of patients. This includes, but is not limited to, heart rate, glucose levels, and blood pressure in real time for patients both in and out of hospitals and other medical establishments. Research indicates that smart health monitoring systems have enhanced chronic disease management for patients by allowing constant tracking of patients' care and early proactive treatment for the most critical cases (Kumar, et al., 2023). Based on a study, there has been a reduction in emergency admissions of patients by about 20%, and there has been an increase in the adherence of treatments for patients by 30% (Kumar, et al., 2023). Research on AI-driven wearable technologies demonstrates that machine learning algorithms can assess the data from IoT devices to be able to anticipate decline in patient conditions, and limit emergency hospital visits (Xie, et al., 2021). Moreover, IoT devices served as an instrumental tool for telemedicine, especially for healthcare in a post-pandemic era, where patient monitoring was done from home to reduce strain and enhance productivity for hospitals and patient access (Kelly, Campbell, Gong, & Scuffham, 2020).

IoT technologies are also enhancing hospital efficiency. From workflows driven by automation, smart hospital beds for patients, and predictive resource management. Research indicates that IoT-powered hospital management systems can improve workforce distribution and track various medical equipment in real time, which contributes to the increase in hospital efficiency (Ahmad, 2024). It was demonstrated in a study that the implementation of these hospital systems and networks have reduced the response times of hospital workers and enhanced the proficiency of room allocation to maintain a proper flow (Sujith, Sajja, Mahalakshmi, Nuhmani, & Prasanalakshmi, 2022). Also, through predictive analytics, these smart devices are capable of anticipating equipment failures and/or patient surges, which helps hospitals with allocating their resources more effectively (Bagheri, et al., 2024).

Regardless of these developments, data security and patient privacy are still concerns that pose as barriers for complete adoption of IoT in healthcare. Sensitive health data is constantly flowing through IoT networks, which makes them more susceptible to cyberattacks, leading to an increase in concerns about data breaches in many healthcare networks and unauthorized access (Kumar, et al., 2023). Additionally, there are compatibility challenges between different IoT healthcare devices, which can restrict data integration through many healthcare systems and affect the efficiency of both the devices and the hospital workflow (Kelly, Campbell, Gong, & Scuffham, 2020). But solutions include blockchain technologies and encrypted authentication protocols that are being developed and used for the security of patient data and addressing compatibility issues (Kumar, et al., 2023).

The Intersection of AI and IoT in Healthcare

The integration of both Artificial Intelligence (AI) and Internet of Things (IoT) has become a cutting-edge development in healthcare through a combination of real-time data management and AI-driven data analytics for the improvement of patient monitoring, diagnostics, and hospital efficiency. Based on my research, AI improves the IoT data gathered from the many interconnected systems, the significance of smart hospitals that consist of AI and IoT integrations, and the challenges of system compatibility and security risk through many algorithms (Bagheri et al., 2024; Ahmad, 2024). These improvements generate challenges, such as compatibility between systems, cybersecurity risks, and data standardization, which is the constant formatting of data (Kumar, et al., 2023).

A prominent advancement in the intersection of AI and IoT is the capability to monitor patients remotely, both in hospital settings and beyond. AI algorithms work very hard to analyze the data it is given from wearable IoT devices to detect any form of health issues/decline in patients from chronic conditions such as diabetes and heart disease (Xie, et al., 2021). Research shows that there has been a 25% decrease in emergency hospital visits from the combination of this wearable technology and automation systems (Xie, et al., 2021).In smart hospitals, there are many implementations of AI and IoT combinations to enforce medical automation and predictive patient resource management while also managing the concerns of data security and the data that is constantly being managed (Ahmad, 2024).

Conclusion

Artificial Intelligence (AI) and the Internet of Things (IoT) in healthcare have greatly enhanced patient care, diagnostics, and the ongoing operations of hospitals. We have early disease detection for patients so that their health concerns can be managed more efficiently and quickly. For AI-powered analytics there are personalized treatments unique to almost every patient in and out of the hospital. And there are predictive healthcare methods that these technologies provide as well. For IoT devices, they focus on remote monitoring for patients and real-time data management. There are also automation implementations to increase workflow. Both of these technologies create further advancements to the overall success and management of the healthcare industry. But, there are risks of compatibility issues for different smart devices operating on one network, cybersecurity concerns where patient data is vulnerable to threats and breaches, and regulatory issues where there may be constraints placed on the growth of the adaptations from local governments.

Future Direction in AI and IoT in Healthcare

The future for Artificial Intelligence and the Internet of Things in healthcare will primarily focus on how the security of these sectors in technology can improve security and strengthen compatibility between systems. One barrier that is present is the lack of standard frameworks to enforce the seamless flow of communication and data, and researchers are working constantly to accomplish this goal (Zikria, Ali, Afzal, & Kim, 2021). There is also a goal to reduce bias in AI algorithms; there are ethical concerns resulting from the unfair outcomes for underrepresented groups and minorities from these algorithms, while these experts are working to better their health detection systems (Xie, et al., 2021). The questions that remain are how there can be a balance between equity, privacy, and accessibility as these systems grow and become more advanced.

References

Ahmad, A. (2024, August). *Enhancing Hospital Efficiency through IoT and AI: A Smart Healthcare System.* Retrieved from ResearchGate Website: https://www.researchgate.net/publication/384012959_Enhancing_Hospital_Efficiency_through_IoT_and_AI_A_Smart_Healthcare_System

Bagheri, M., Bagheritaba, M., Alizadeh, S., Salemizadeh Parizi, M., Matoufinia, P., & Luo, Y. (2024, June 12). *AI-Driven Decision-Making in Healthcare Information Systems: A Comprehensive Review.* Retrieved from Google Scholar Website: https://www.preprints.org/manuscript/202406.0790/download/final_file

Bajwa, J., Munir, U., Nori, A., & Williams, B. (2021, July). *Artificial intelligence in healthcare: transforming the practice of medicine.* Retrieved from National Library of Medicine: https://pmc.ncbi.nlm.nih.gov/articles/PMC8285156/

Elendu, C., Amaechi, D. C., Elendu, T. C., Jingwa, K. A., Okoye, O. K., John Okah, M., . . . Alimi, H. A. (2023, December 15). *Ethical implications of AI and robotics in healthcare: A review.* Retrieved from Medicine Journals Website: https://journals.lww.com/md-journal/fulltext/2023/12150/ethical_implications_of_ai_and_robotics_in.101.aspx

Hou, X., Shen, G., Zhou, L., Li, Y., Wang, T., & Ma, X. (2022, March 10). *Artificial Intelligence in Cervical Cancer Screening and Diagnosis.* Retrieved from Frontiers Website: https://www.frontiersin.org/journals/oncology/articles/10.3389/fonc.2022.851367/full

Jimma, B. L. (2023). *Artificial intelligence in healthcare: A bibliometric analysis.* Retrieved from ScienceDirect Website: https://www.sciencedirect.com/science/article/pii/S2772503023000014?via%3Dihub

Kelly, J. T., Campbell, K. L., Gong, E., & Scuffham, P. (2020, November 10). *The Internet of Things: Impact and Implications for Health Care Delivery.* Retrieved from National Library of Medicine Website: https://pubmed.ncbi.nlm.nih.gov/33170132/

Kumar, M., Kumar, A., Verma, S., Bhattacharya, P., Ghimire, D., Kim, S.-h., & Hosen, A. S. (2023, April 28). *Healthcare Internet of Things (H-IoT): Current Trends, Future Prospects, Applications, Challenges, and Security Issues.* Retrieved from MDPI Website: https://www.mdpi.com/2079-9292/12/9/2050

Nagavelli, U., Samanta, D., & Chakraborty, P. (2022, February 27). *Machine Learning Technology-Based Heart Disease Detection Models.* Retrieved from Wiley Online Library Website: https://onlinelibrary.wiley.com/doi/full/10.1155/2022/7351061

Sujith, A. V., Sajja, G. S., Mahalakshmi, V., Nuhmani, S., & Prasanalakshmi, B. (2022). *Systematic review of smart health monitoring using deep learning and Artificial intelligence.* Retrieved from ScienceDirect Website: https://www.sciencedirect.com/science/article/pii/S2772528621000285?via%3Dihub

Xie, Y., Lu, L., Gao, F., He, S.-j., Zhao, H.-j., Fang, Y., . . . Dong, Z. (2021, December 24). *Integration of Artificial Intelligence, Blockchain, and Wearable Technology for Chronic Disease Management: A New Paradigm in Smart Healthcare.* Retrieved from Springer Nature Link Website: https://link.springer.com/article/10.1007/s11596-021-2485-0

Zikria, Y. B., Ali, R., Afzal, M. K., & Kim, S. W. (2021, February 7). *Next-Generation Internet of Things (IoT): Opportunities, Challenges, and Solutions.* Retrieved from MDPI Website: https://www.mdpi.com/1424-8220/21/4/1174

Shemar Anglin ENGL394-9012 Sajeela Yaqub

Word Count: 2344

Citation Style: APA7

Target Journal: Journal of Young Investigators